

Sonic Firmware Documentation V64

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 Many thanks to Sonic Team : Zylka, Kolyan, Toni, Yves, Francois and Adam.

This software is a fork of Armel F4HWN firmware with code from NTOIVOLA, EGZUMER and DUAL TACHYON.

EXPLANATION OF BUTTON SYMBOLS:

1 – Simple press.

L1 – Long press.

F1– Press after the F button.

MR MAIN SCREEN (CHANNELS) | VFO (FREQUENCY)

L1 – (MR) – Transfer the channel frequency to VFO mode.

L3 – Change the MR/VFO mode on the selected channel.

L4 – (MR and VFO) – Bandwidth change.

L5 – (MR) – Change the list number for the channel. (VFO) – Change of step.

L6 – (MR & VFO) – Power Change (L/M/H).

L7 – (MR and VFO) – On/Off flashlight indication when receiving a signal.

L9 – (MR and VFO) – On/Off constant backlight.

L0 – AM/FM/USB Modulation Change

F2 – (MR) – Delete Channel. (VFO) – Preserve the frequency.

F6 – Spectrum analyzer in scanrange mode.

F7 – Spectrum analyzer in scanlist mode.

F8 – Band Spectrum Analyzer.

F9 – Spectrum analyzer in frequency mode.

F0 – Radio broadcasting (FM radio).

SK1 and **SK2**—Side keys, assignable buttons.

<= and **=>** – Navigation buttons.

MA – Menu.

EXIT – Exit.

#F – Functional "F".

OPERATION IN SPECTRUM MODES

In this mode the scan speed is around 300 ch/s.

The scan is interlaced: the step is minimal 25k and the spectrum is swept with different start frequencies to cover all the frequencies.

For example 125.100, 125.125,... then 125.10833,125.133333...then 125.11666,125.3666...

This to improve detection.

Different types of spectrum are called by the quick buttons

F6 - Range, Freq start, stop and step set in **[5]** menu.

Scans between 2 selected frequencies with the given step

F7 - Scanlists

Scans the frequencies in scanlists that can be defined in VFO or Chirp*.

50 lists are available with their name.

F8 - Bands

Scans ranges that are defined in bands, can be defined in Chirp* or in a channel greater than 974 for Sonic 1K / 4046 for Sonic 4K.

Channel Frequency is START, Offset + is STOP, STEP is step.

F9 - Frequency: spectrum around VFO frequency, 128 frequencies with current step.

*See Chirp section below.

- 1 – Skip the frequency in the scan mode.
- 2 – Turn on/off the backlight).
- 3 – Bandwidth in the spectrum. In the options menu, change the value.
- 4 – List of scanned lists or bands.
- 5 – Options menu. Navigation through the menu < and >. Change the values with buttons 1 and 3.
- 6 – Changing the modes of the spectrum analyzer Scanlists / Bands.
- 7 - Save Settings. Save after configuring and selecting the lists and the options will be applied once enabled.
- 8 – Change the display screen between Classic spectrum, Smooth spectrum and SCAN.

Smooth Spectrum



Classic Spectrum



9 – Change of AM/FM/USB modulation.

0 – Switch to the history mode. Displays the RX frequencies, channels, code and time in s.

* and # DYNSQL **Level setting** - RSSI spike threshold for instant peak capture. For weak and fuzzy signals,

increase or disable.

L* and **L#** DYNSQL OFF This will fill the history, but won't stop.

SK1 – Frequency Lock (FL) then monitor and back to normal mode.

FL locks the frequency, opens listening on receive

Monitor, keeps the frequency locked and open listening

SK2 – Blacklist a frequency, it is visible in history with a **#**

OPERATION IN ULTRA WATCH MODE

In this mode the scan speed is around 600 ch/s, in spectrum it is around 300ch/s
Keys are the same as for spectrum.

UW in Scanlist mode



Frequency is displayed in big

Received DCS and CTSS codes are displayed in small

RX displays last received channel.
RX➤ when receiving

in PTT Channel mode:

TX to show current selected channel, change with <> keys into the selected scanlists

In PTT LastRx mode:

TX shows last received channel, ready for TX.

TX+ or TX- when an offset is used

TX➤ when transmitting with PTT

UW Band mode



Frequency is displayed in big

DCS and CTSS codes are displayed in small

Bx shows current band

Here we receive so RX> is displayed with an smeter below.

TX shows the last received frequency, ready to use for TX. In band mode, no channel setting is used.

Keys in history

This mode lists received signals, with a time counter. Up to 200 items are stored.

<,> - Move to previous / next item, automatically set to frequency lock to the selected item

1 – Scans history entries.

2 – Saves the frequency to a free channel (The first empty found).

3 – Delete the entry.

5 – History menu

Clear history : erase the history

Clear blacklist – Clears all black listed entries.

Clear non blacklist - Clears all non black listed entries.

Save history – Saves the history.

Save hist to ch : Saves all the history items to free channels. Tip: try to avoid holes.

6 – Launch close call: detection of strong signals. When found, it is stored in history and close call continues.

8 - Save history to flash.

EXIT : go back to UW or SPECTRUM

Keys in band selection

<,> - Move to previous / next item

1,3 – Select a preset

4 – Activate or deactivate one band.

5 – Activate one band and deactivate any other

7 – Save current preset.

MENU - activate selected item and start scanning

EXIT : go back to SCAN or SPECTRUM

Keys in scanlist selection

<,> - Move to previous / next item

4 – Activate or deactivate one scanlist.

5 – Activate one scanlist and deactivate any other

MENU - activate selected item and start scanning

EXIT : go back to SCAN or SPECTRUM

PARAMETERS OF THE SPECTRUM ANALYZER

SPECTRUMDELAY – Pause time after signal detection (in ms) – if 0, immediately resumes scanning.

MAXLISTENTIME – Maximum time to listen to the signal (in seconds) – after that, the scan continues.

FSTART – Initial frequency of the scan range (in MHz).

FSTOP – End frequency of the scan range (in MHz).

STEP – Frequency scan step (in kHz): 0.01; 0.05; 0.1; 0.5; 1.0; 2.0; 5.0; 6.25; 10.0; 12.5; 20.0; 25.0; 50.0; 100.0.

LISTENBW – Bandwidth: **Wide** (25 kHz) / **Narrow** (12.5 kHz)/ **AIR** (8.33kHz).

MODULATION – Modulation type: **FM/AM/USB**.

POWERSAVE – Power saving mode: Off /200ms /500ms /1s /2s /5s.

KEY UNLOCKED – THE TIME BEFORE THE KEYPAD IS LOCKED.

NOIS LVL OFF – Noise level for disabling reception (above this – the signal is counted).

POPUPS – Popup display time (in ms) – 0 = disabled.

RECORD TRIG – An additional threshold for recording in history. The higher the threshold, the stronger the signal will be recorded.

SOUND BOOST – Amplify sound. Increase bass frequencies.

PTT – What to do when pressing PTT: LAST VFO /NINJA /LAST RECEIVED/AUTO ROGER .

- NINJA chooses a random frequency in the current scanlist.
- AUTO ROGER :Choose your Roger in vfo menu, select a frequency or a channel in vfo, in spectrum select ptt auto Roger in the menu, it will send a Roger beep every duration you select.

MONITOR SL – Includes the "MON" priority list in all scan lists.

RESET DEFAULT – Reset all spectrum parameters to factory settings.

DETAILED MONITORING IN SPECTRUM MODE VIA THE "MENU" BUTTON

Move through the low-level register menu with buttons **2** and **8**. Change values with the **<=** and **=>** buttons

LNAS – Low Noise Amplifier (LNA) – signal amplification stages in front of the mixer.

LNA – Gain of the first LNA stage – higher = stronger signal but more interference.

PGA – Programmable Gain Amplifier (after LNA). – Balance between sensitivity and distortion.

MIX – Mixer Gain (Frequency Mixing) – Higher = Better Dynamic Range but More Noise.

XTAL F MODE SELECT – Quartz filter (XTAL) operating mode. Filtering (reduces noise, but can distort).

OFF AF RX DE-EMP – Disable demphasis (pre-distortion correction) during reception.

GAIN AFTER FM DEMOD – Adjusts the reception volume.

RF TX DEVIATION – Transmission frequency deviation (Fm deviation) – affects the transmission bandwidth (higher = louder).

COMPRESS AF TX RATIO – Compression ratio of audio during transmission – compresses the dynamic range.

COMPRESS AF TX 0 DB – Compression level 0 db during transmission – the compression start point.

COMPRESS AF TX NOISE – Noise reduction in quiet areas.

MIC AGC DISABLE – Disable automatic microphone gain control (AGC).

AFC RANGE SELECT – Operating range of AFC (Automatic Frequency Tuning).

AFC DISABLE – Disable AFC.

AFC SPEED – AFC Speed.

OPERATION IN BROADCAST RADIO MODE "FM"

The firmware has a simplified FM radio. Use **the <= and => buttons** to search for the nearest available broadcasting frequency in automatic mode. The * button will change the **Auto mode** to **manual** search.

Press key 1 to 6 to select the corresponding memory.

Long press key 1 to 6 to save a frequency to that memory.

Press 0 to toggle receiving the VFO frequency, RX MON displayed or removed.

MAIN MENU

THERE IS NO HIDDEN MENU IN THE FIRMWARE

1.	Step	The frequency step (in kHz)
2.	SQL	The level of noise cancellation (Squelch) is from 0 (always open) to 9 (very strict). Set for both bands (A and B)
3.	Mode	Demodulation mode, default - FM, AM and USB can only be used for listening
4.	RxDCS	Setting Digital Coded Squelch, if enabled, the squelch will only open after receiving that subtone, applicable for FM modulation only.
5.	RxCTCS	Setting an analog subtone to reception (Continuous Tone Coded Squelch), if the function is enabled, the noise canceler will open only after receiving this subtone, applicable only for FM modulation.
6.	TxDCS	Setting the digital encoded digital subtone to transmit, if the function is enabled, the transceiver will transmit this tone, applicable only for FM modulation.
7.	TxCTCS	Setting the analog subtone to transmit, if the function is enabled, the transceiver will transmit this tone, applicable for FM modulation only.
8.	TxOffs	Transmitter frequency offset value in MHz, transmit/receive at different frequencies.
9.	TxODir	Direction (+/-) of transmitter frequency offset when transmitting/receiving at different frequencies
10.	W/N	The bandwidth used by the transceiver (WIDE (25 kHz), NARROW (12.5 kHz).
11.	SetNFM	Sets Narrow FM to Narrow or Narrower
12.	Scramb	Selects Scrambling type
13.	Compnd	Compander - Soft audio filter for clearer sound. Amplifies the received desired signal even when the signal is weak at the transmitting microphone, improving the signal-to-noise ratio. To work, it must be turned on on both walkie-talkies.
14.	Roger	On/off "roger beep" (end-of-pass tone).
15.	STE	Noise suppressor, eliminates noise at the end of the transmission. When PTT is released, a short 55Hz tone is emitted, signaling to other transceivers to mute
16.	RP STE	Noise Cancelling Tail Eliminator Repeater
17.	ChSave	Saves the frequency to a chosen memory
18.	ChDele	Deletes a channel memory
19.	ChName	Name of the memory channel.
20.	F1Shrt	Function assignment by short pressing the F1 side key.
21.	F1Long	Function assignment when the F1 side key is pressed for a long time.
22.	F2Shrt	Function assignment by short pressing the F2 side key.
23.	F2Long	Assign a function when you press the F2 side key for a long time.
24.	M Long	Assign a function when you long press the M key
25.	KeyLck	Automatically lock the keypad after a set time
26.	TxOut	Transmit Timeout - The amount of time that PTT can be held during transmission (in seconds). Protection against accidental clamping of PTT.
27.	BatSav	Battery Save: OFF / 1:1 / 1:2 / 1:4 / 1:8.
28.	Mic	Microphone sensitivity (microphone amplifier).
29.	POnMsg	Enable/disable the display of a message when turned on (Power On Message).
30.	BLTime	Screen backlight time when buttons are pressed.
31.	BLMin	Minimum backlight brightness (in percentage).
32.	BLMax	Maximum backlight brightness (in percentage).
33.	BLTxRx	Transmit/Receive Backlight: OFF/TX/RX/TX+RX.
34.	SetInv	Enable walkie-talkie screen color inversion
35.	BtnInv	Reverse the direction of the UP and DOWN buttons
36.	SetLck	Setting the keypad lock type with or without PTT.
37.	BatTxt	Battery Display Type, %, V or disabled.
38.	MicBar	Enabling/disabling the microphone bar
39.	SetTmr	Set the display of RX/TX timers on the screen
40.	F Lock	Setting up Frequency Lock. What frequencies are allowed or prohibited to work on.

41.	BatCal	Battery Calibration.
42.	BatTyp	Choice of battery type (Li-ion, NiMH, etc.) and capacity.
43.	Reset	Channel Preservation Reset (NOCH) or Full Factory Reset including Saved Channels (ALL)

Installation

RS232 version uses the kenwood cable.

USB version uses USB-C cable.



RS232 Kenwood cable



USB-C cable

use the same procedure as F4HWN, after that :

- If necessary, you can do a full reset of the radio. press **the PTT** and **Fn2 buttons** , turn on the radio. A warning message will be displayed on the screen. hold the buttons for 4 seconds before resetting.

In case of issues to install, try to first install N7SIX firmware to reset the memory.

CHIRP Driver

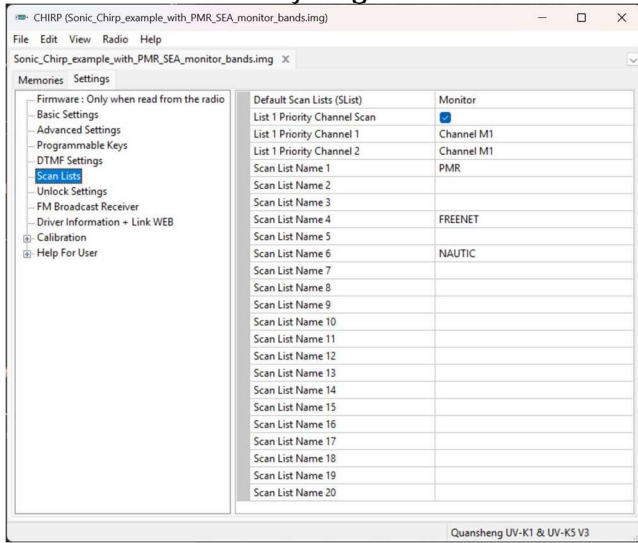
Chirp_Sonic.py in [DOWNLOAD HERE](#)

This driver also allows

- customization of the 50 bands on channels 976 to 1024.
Select start frequency and offset as stop frequency, step and modulation

Chirp Configuration

- It is necessary to give names to the scanlists like so:



- Memory channels 975 to 1024 store custom band settings for **Sonic 1K**.
- Memory channels 4047 to 4096 store custom band settings for **Sonic 4K**.
- For each, define:
 - Band Name.
 - Start and End frequency (via offset).
 - Modulation & Frequency Step.

[illegible]

- Scanlist Monitor setting Select up to 20ch to Monitor scanlist anywhere in the channels.

[illegible]

Flash mapping considerations:

Here is a very simplified view of the regions in Flash memory:
Sonic 1K uses his slot, or the one set with option SetCfg for his parameters.
Sonic 4K uses a fixed region for his channels. If you install multiple Sonic 4K they will share the same region.
A Sonic 8K is possible, but it would be conflicting with F4HWN Labs apps region.
When first installing Sonic it is recommended to RESET ALL with PTT+SK2 at boot.
LOGO and CALIBRATION in Sonic are not available from UvStudio, please use F4HWN to manage them.
The RESET feature is not available in UvStudio.
The apps are not available in Sonic.

